

Beamer⁺

Interactive Presentations

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Beamer⁺ is a lightweight web app that enhances static slides with embedded media, real-time drawing tools, intuitive navigation, and a suite of powerful presentation features.

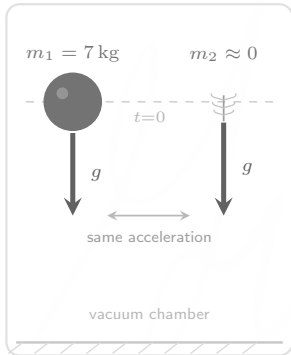
using Beamer⁺

follow these steps to use Beamer⁺

- (1) **create**
create slides using any standard tool (including L^AT_EX)
- (2) **configure**
write `config` files to specify the placement and behaviour of any embedded media
- (3) **package**
combine the slides, embedded media, and `config` files into a package
- (4) **present**
open the package with Beamer⁺, which will display the slides with all embedded media integrated seamlessly

videos

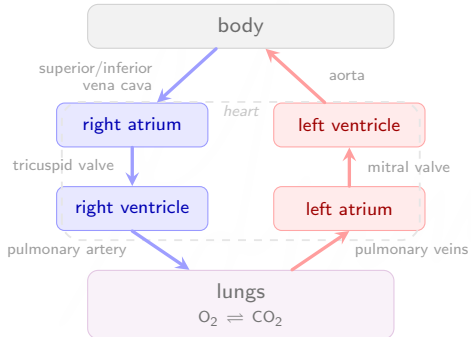
supported formats include .mp4, .mov, .mpeg



gravitational acceleration is
independent of mass

models

supported formats include .glpb



blood flow through the heart
(deoxygenated blood / oxygenated blood)

audio

supported formats include .mp3, .wav

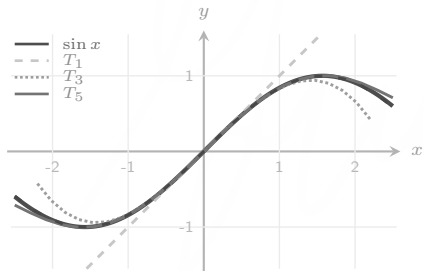
custom widgets

supported formats include `.html`

Any smooth function can be approximated by polynomials.

A function f can be approximated around $x_0 = 0$ as follows:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$



step	approximation
T_1	x
T_3	$x - x^3/3!$
T_5	$x - x^3/3! + x^5/5!$
T_5	$x - x^3/3! + x^5/5! - x^7/7!$
$\vdots$	$\vdots$

expansion of $f(x) = \sin(x)$ around $x_0 = 0$

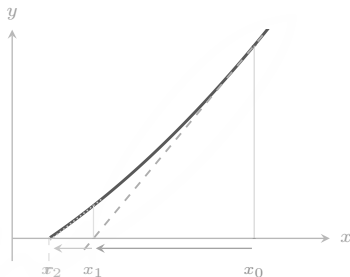


Newton's method finds roots iteratively by following tangent lines.

Given $f(x) = 0$ with initial guess x_0 , iterate:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Each step replaces f by its **tangent line** at x_n and takes the root of that line as the next guess.



using Newton's method to solve $x^2 - 2 = 0$
from $x_0 = 2$ ($x = \sqrt{2}$)

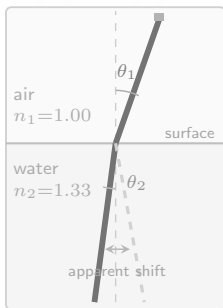


Light bends when it crosses the boundary between two materials.

At the edge between two media, with refractive indices n_1 and n_2 :

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

where θ_1 is the angle of incidence, and θ_2 is the angle of refraction.

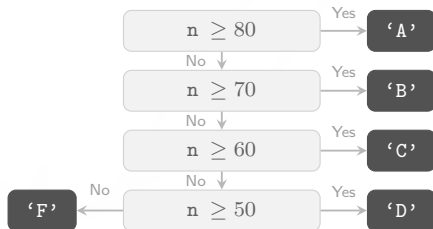


refraction of a pencil in a glass of water



Complete the function, `letter_grade` that takes as input a student's numerical grade, `n`, and returns its corresponding letter grade:

numerical grade, n	letter grade
$n \geq 80$	A
$70 \leq n \leq 79$	B
$60 \leq n \leq 69$	C
$50 \leq n \leq 59$	D
$n \leq 49$	F



numerical to letter grade conversion (left) and algorithm flow chart (right)















learn more

visit the link below to learn more

`chandra-gummaluru.github.io/post/beamer-plus`